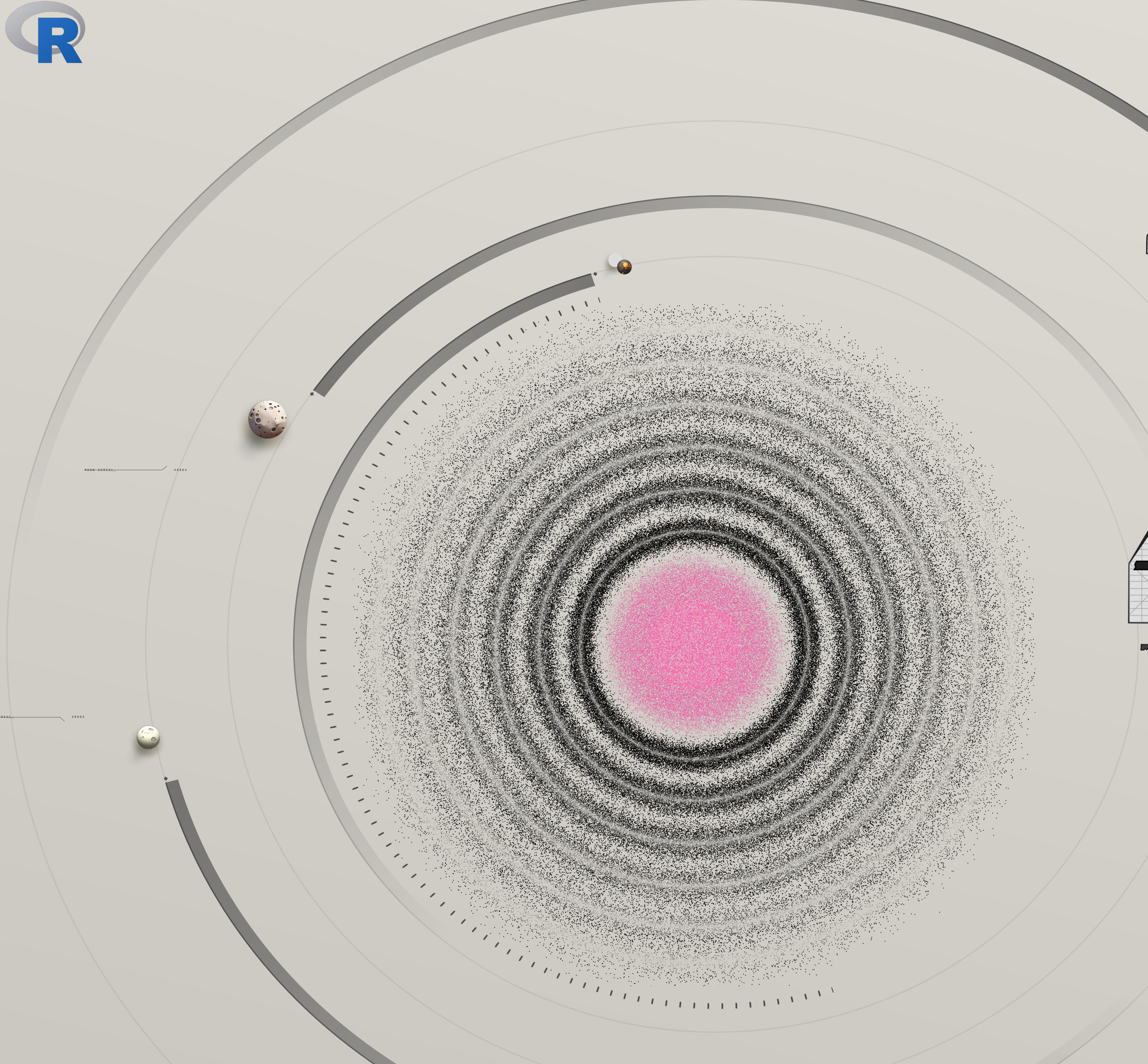


M A V E N S P A C E

C H A L L E N G E

BY DEV VERMA

This project is built on Jonathan McDowell's Satellite Launch Log (JSR) - widely recognized as the gold standard for verified orbital launch data. After cleaning and deduplicating over 27,000 records, the verified dataset reveals 6,617 unique orbital launches (1957–2024) forming the factual backbone of the analysis. To enrich the verified JSR archive with modern context, additional metadata from Maven, SpaceX, and The Space Devs APIs was integrated - connecting verified history with current and projected activity. The result merges factual precision with visual storytelling: a data-journalism view of humanity's ascent beyond Earth, inspired by the design philosophy of David McCandless - clarity, beauty, and truth through data.



Cosmic Swarm | Reading Guide · 1957 → 2024 (Time radiates outward from center)

The **PINK** nucleus marks Plesetsk Cosmodrome (Russia), representing 1,670 of 6,617 verified orbital launches (25.2%) in the JSR-derived dataset (1957–2024).

Each dot corresponds to one verified orbital launch, forming a radial timeline of humanity's ascent into orbit.

The rings trace time radiating outward:

- Core: early Space Age — Sputnik 1, Vostok, and the first U.S. flights
- Mid-bands: Cold War proliferation and early commercial activity
- Outer bands: the modern reusable-rocket era — Falcon 9, Soyuz-2, Long March 5, and new entrants

Visual note:

The underlying layout uses one point per verified launch (6,617 points).

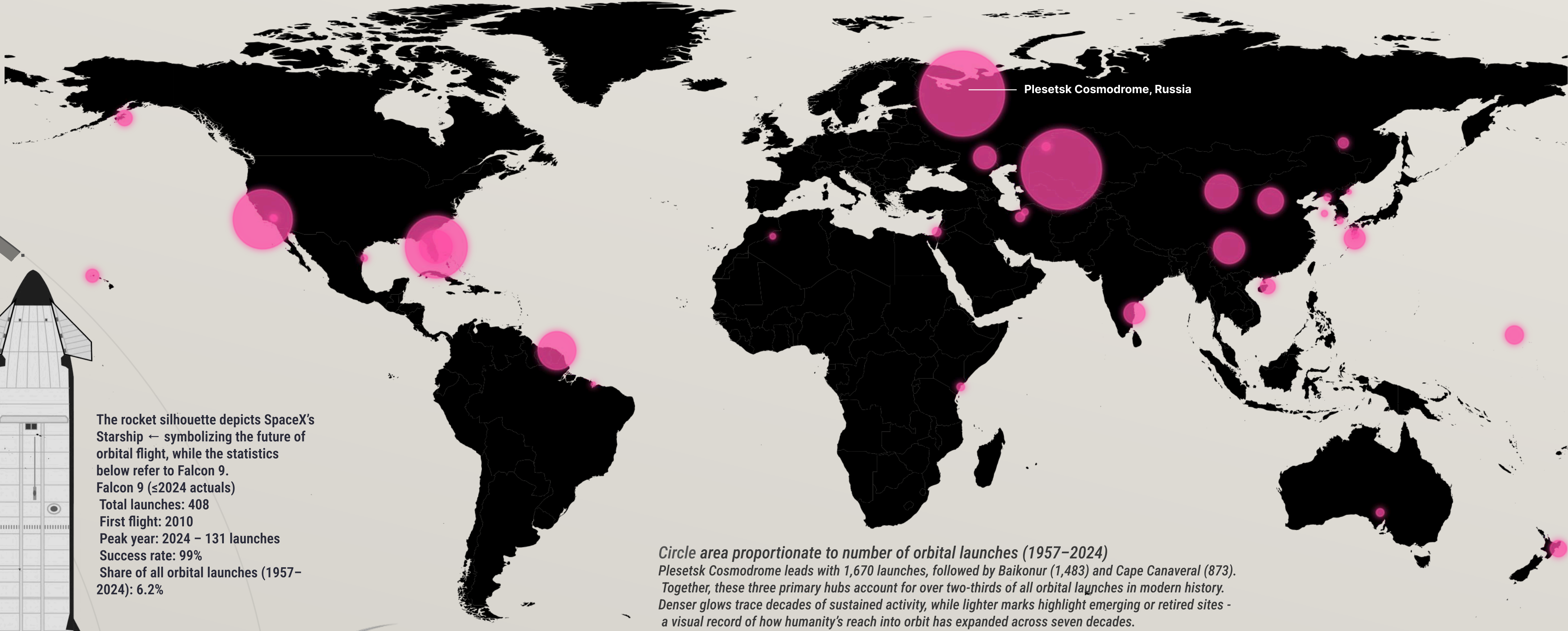
For print clarity, a subtle grain layer enhances the concentric halo and makes decade transitions more readable.

Data source & reproducible code:

GitHub · jsr-orbital-launch-history-1957-2024

LIFTOFF ATLAS | Humanity's Global Launch Footprint (1957 – 2024)

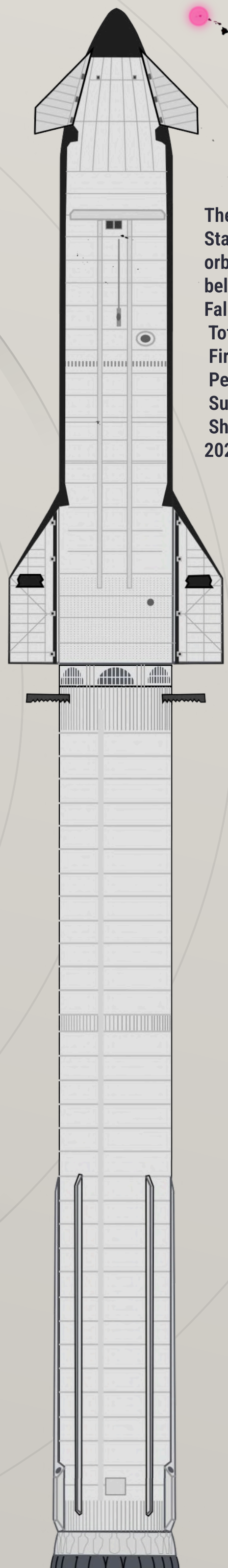
6,617 orbital launches (1957–2024)
92.6% successful | 6.2% failed | 1.2% partial / other.
Source : Jonathan's Space Report (JSR 1957 – 2024) | Deduped unique orbital launches



The rocket silhouette depicts SpaceX's Starship — symbolizing the future of orbital flight, while the statistics below refer to Falcon 9.

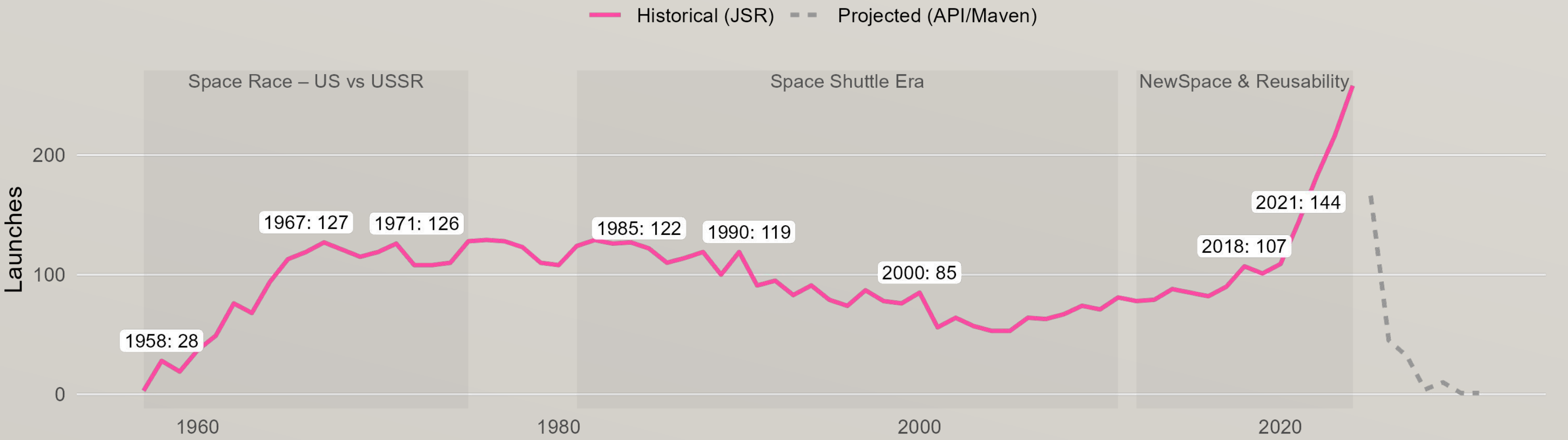
Falcon 9 (≤2024 actuals)
Total launches: 408
First flight: 2010
Peak year: 2024 – 131 launches
Success rate: 99%
Share of all orbital launches (1957–2024): 6.2%

SPACEX



Humanity's Orbital Launches per Year (1957–2031)

Solid: JSR-verified launches (1957–2024) • Dashed: projected missions from Maven, SpaceX & The Space Devs (2025–2031)



SpaceX | Orbital Launches per Year (2006–2024)

From single-digit launches to record cadence • JSR-verified data

